When do field gradients pay off in motor design? Forward-model validation for a controlled optimization study

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Abstract

Evaluating the benefit of field gradients in motor design requires a forward model that resolves the quantity being optimized. We build a nonlinear finite-element reproduction of a published permanent-magnet motor and find that agreement in average torque does not imply agreement in torque ripple. A consistent-tangent Newton solver, tabulated iron law and rotating sector model reproduce the initial design's loaded mean torque within 0.16% on the finest of four meshes. The population standard deviation of torque differs by 9.9% and 11.3% at half and full current, while zero-current cogging ripple is 29.1% lower. These discrepancies exceed an empirical spatial-refinement estimate. Material-endpoint, solver-tolerance and winding-area checks do not explain the gap; differences in interface coupling and the reference discretization remain unresolved. We provide the model, raw torque records and a comparison protocol that separates useful shape freedom from derivative cost. The forward study is complete at the reported scope; motor shape-gradient verification and matched optimization timing remain prospective. No optimization speedup is claimed.

1 Introduction

Average torque can agree closely between two motor models while the quantity used to optimize the design still disagrees. In our reproduction of the initial motor of Wiesheu et al. (2024), loaded mean torque differs by at most 0.16%, but torque ripple differs by about 10–11%. This matters for a study of optimization cost: reaching a target quickly has little meaning if the target changes with the field model.

The larger question is when it is worth computing derivatives of a full finite-element model. Adjoint methods can obtain derivatives with respect to many design variables without a separate field solve for each variable. That computational property does not by itself establish a practical advantage over a less expensive physical model or a calibrated surrogate. The additional design directions must affect useful performance, and their benefit must survive numerical refinement and common physical constraints. Nonlinear slotted motors provide a setting in which local geometry, saturation and rotor position can affect the torque waveform together.

This manuscript establishes the forward-model evidence for that comparison and states the remaining experiment. Its contributions at the present scope are:

- a reproducible reconstruction of the published initial motor, with explicit material, excitation, rotation and torque conventions;
- a nonlinear finite-element model checked through material tests, manufactured solutions, an Ampère-law reference and sector comparisons;

- a four-mesh torque study that separates agreement in mean torque from the unresolved ripple discrepancy; and
- a prospective protocol for comparing gradients, reduced models and derivative-free search at the same validated design target.

The numerical formulation uses established methods. We do not claim a new adjoint principle or a completed motor shape-optimization result. The stronger contribution sought by the ongoing study is a measured account of when additional shape freedom justifies the cost of full-field derivatives.

2 Related work

Wiesheu et al. (2024) combine spline-based rotor and stator shape optimization with engineering parameters for a nonlinear permanent-magnet synchronous motor. Their public implementation and saved outputs provide the reproduction target here. Their related study compares parameter and shape optimization with isogeometric analysis (Wiesheu et al., 2025). We inherit the initial geometry, material table and excitation conventions; we replace the field discretization and rotor–stator coupling. The resulting comparison is between numerical models, rather than a comparison with a measured physical motor.

Differentiable motor design is established prior work. Scotzniovsky et al. (2024) combine geometric deformation, mesh motion and electromagnetic analysis with adjoint-based derivatives. The existence of a differentiable implementation is therefore insufficient as a novelty claim. A useful comparison must establish what designs can be reached, how accurately they are evaluated and how much computation is required.

Inexpensive alternatives also need substantive treatment. Kemmetmüller et al. (2015) use nonlinear magnetic equivalent circuits for torque control, including saturation and nonsinusoidal behavior. Their control problem differs from shape optimization, but it demonstrates why saturation alone does not establish the need for a full-field model. Farshbaf Roomi et al. (2022) develop a waveform-targeting surrogate for electrical machine design, while Wen and Zahr (2023) adapt reduced models along a shape-optimization trajectory in fluid examples. These approaches motivate waveform-aware and adaptive baselines. Their published performance is not transferred to the motor studied here.

3 Motor model and reconstruction

3.1 Geometry and operating states

The full machine has six poles and 36 stator slots. We model a 60° sector, occupying $60° \le \theta \le 120°$ at zero rotation, with antiperiodic conditions on the radial boundaries. The rotor inner and outer radii are 16 and 44 mm; the stator inner and outer radii are 45 and 67.5 mm. Axial length is $L = 0.1\,\mathrm{m}$. The 1 mm air gap is split at the coupling radius $r_c = 44.5\,\mathrm{mm}$. Figure 1 shows the material-tagged reconstruction.

The reconstruction uses the initial NURBS exports in the public snapshot. Their control points are stored with weights already multiplied in, and patch orientation must be accounted for when integrating areas. Two thin rotor patches described as air slits in source comments receive iron under the reference's material assignment; we follow that assignment. The mesh generator preserves the winding groups separately, since their two characteristic areas are approximately 36.29 and $31.22\,\mathrm{mm}^2$.

At mechanical angle β, phase currents are

$$I_q(\beta) = aI_* \sin(\phi_0 + 3\beta + \delta_q), \qquad (\delta_U, \delta_V, \delta_W) = (0, -2\pi/3, 2\pi/3), \tag{1}$$

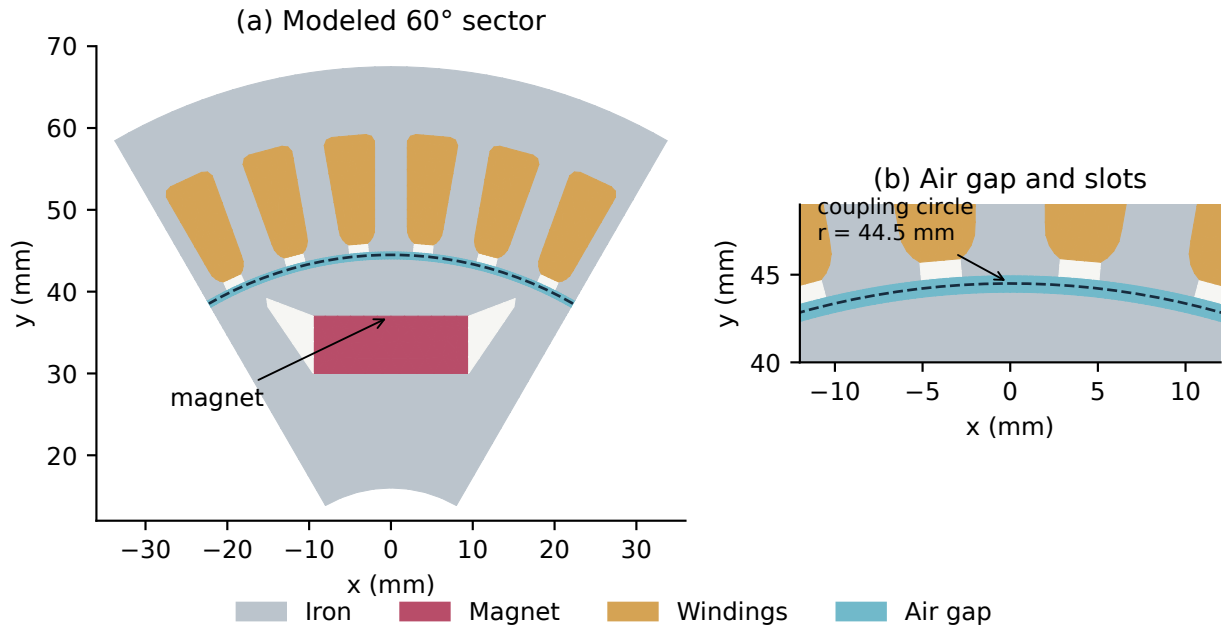


Figure 1: Initial-design geometry rendered directly from the committed coarsest mesh. The dashed line is the rotor–stator coupling circle. The right panel enlarges the gap and slot openings. Material boundaries are shown without the element edges; the geometry is not an optimized result.

where $a \in \{0, 0.5, 1\}$, $I_* = 9.677657374\,\mathrm{A}$, and the electrical phase offset $\phi_0 = 0$ for the initial design. Angles in Eq. (1) are in radians. Each winding group carries current density $J_g = 12 s_g I_q / |\Omega_g|$, with its own area and prescribed sign $s_g \in \{-1, 1\}$. Using a common area would change the current density between groups by about 16%. We compare 20 distinct mechanical angles, $0°, 1°, \ldots, 19°$, excluding the duplicated endpoint in the source CSV. The reference's optimization grid contains the ten even angles; both grids are retained in the artifacts.

3.2 Nonlinear field equation

The two-dimensional magnetic vector potential $A = A_z$ determines the flux density $\mathbf{B} = (\partial_y A, -\partial_x A)$. Iron is described by an isotropic reluctivity $\nu(|\mathbf{B}|^2)$, where reluctivity is the inverse of permeability. For a test function v that satisfies the homogeneous constraints, the weak equation is

$$\int_\Omega \nu(|\nabla A|^2) \nabla A \cdot \nabla v \, \mathrm{d}\Omega = \int_\Omega J_z v \, \mathrm{d}\Omega + \int_{\Omega_m} \mathbf{M} \cdot (\partial_y v, -\partial_x v) \, \mathrm{d}\Omega. \tag{2}$$

Here $\mathbf{M}$ denotes the permanent-magnet source in $\mathbf{H} = \nu \mathbf{B} - \mathbf{M}$. In the magnet, $\mu_r = 1.05$ and $B_r = 1.4\,\mathrm{T}$, so its source magnitude is $H_c = B_r/(\mu_0 \mu_r)$, not B_r/μ_0. Its direction is $\pi/2$ in the rotor frame and rotates with the rotor. Air and windings have $\mu_r = 1$. The inner and outer circular boundaries have $A = 0$.

We use continuous piecewise-linear triangles. With nodal vector u, element gradients $g_i = \nabla \varphi_i$, $G = \nabla A$ and $s = G \cdot G$, the residual derivative on one element is

$$K^e_{ij} = |\Omega_e| \left[\nu(s) g_i \cdot g_j + 2\nu'(s)(g_i \cdot G)(g_j \cdot G) \right]. \tag{3}$$

The second term is required when the material changes with field strength. Newton steps use this consistent tangent, with damping and a stopping reference independent of the initial guess.

Constant potential offsets, where allowed by the boundary conditions, do not change that reference. The solver reports whether it stopped at the requested tolerance or an estimated floating-point floor. All 500 solves in the recorded motor study stopped at the requested tolerance.

The iron law uses 47 tabulated M27 B–H points from 0 to 2.30 T. Monotonicity is imposed on $H(B)$, not on $\nu(B^2) = H(B)/B$. In fact, the tabulated H/B decreases between 0.05 and 0.10 T. The two material tangent eigenvalues must satisfy

$$\nu > 0, \qquad \nu + 2s\nu' = \frac{\mathrm{d}H}{\mathrm{d}B} > 0. \tag{4}$$

The GEODE and reference interpolants agree to round-off between 0.05 and 2.25 T. GEODE uses an odd cubic near zero and a continuous slope into the vacuum-slope tail above 2.30 T. The reference uses a low-field clamp and a different last-knot slope. The maximum relative difference in reluctivity near zero is 6.66%; the last segment differs in H by at most 2.17×10^{-4} relatively. We test the torque effect of the low-field difference in Section 5.3.

3.3 Rotation and torque

At the modeled angles, the two circular traces have matching nodes after rotation. The sliding interface identifies their potentials, including the antiperiodic sign on sector wrap-around. Writing the full nodal vector as $u = P\hat{u} + u_D$, the reduced residual and tangent are $\hat{R} = P^T R(P\hat{u} + u_D)$ and $\hat{K} = P^T K P$. The rotor geometry and magnet source rotate together; currents use the same mechanical angle in Eq. (1).

We average Maxwell stress over the air gap using the Arkkio expression

$$T(\beta) = \frac{L}{\mu_0(r_2 - r_1)} \int_{\Omega_{\mathrm{gap}}} r B_r B_\theta \, \mathrm{d}\Omega, \qquad (r_1, r_2) = (44, 45) \text{ mm}. \tag{5}$$

Positive torque acts counter-clockwise on the rotor. All torques in this paper include axial length and refer to the modeled sector. We apply no factor of six when comparing with the saved reference arrays, despite a per-length label appearing in the reference evaluation plots.

The reference instead couples rotor and stator through eight angular harmonic orders $3, 9, \ldots, 45$, represented by 16 sine/cosine modes. It obtains torque from the derivative of that coupling with respect to rotation. The interface and torque formulations therefore differ between the two models even when the geometry and excitations agree.

4 Verification and experiment design

The implementation is checked before the motor comparison. With constant material laws it agrees with the linear assembly to approximately 2×10^{-14} in potential. Central differences of the nonlinear residual agree with the tangent action to below 10^{-9} at favorable step sizes on the synthetic material tests. A manufactured solution gives fitted spatial rates 1.002 in the H^1 seminorm and 2.007 in L^2. For a saturated iron annulus carrying enclosed current I, the independent radial field relation is $H(r) = I/(2\pi r)$, followed by inversion of the material law. The M27 test reaches a mid-annulus relative field error of 3.76×10^{-4} on its finest mesh. These are verification results for the forward model; they do not verify motor shape derivatives.

Sector tests compare the constrained solution with a full-machine solve, check the reduced tangent by finite differences, and exercise positive, negative and wrapped rotation steps. Clockwise and counter-clockwise triangle orderings give equivalent magnet sources and fields. The motor regression also detects changes in magnet strength, magnet direction, phase ordering and rotation direction. Detailed logs accompany the code.

The motor study uses four meshes, L0–L3, with 3,153, 9,937, 36,083 and 139,268 nodes. The coupling circle has $120\,2^{\ell}$ segments at level ℓ, so integer-degree rotation is represented at every level. Each level contains 60 baseline solves: three current levels and 20 angles. A solution at one angle initializes the next angle at the same current. We report the mean and population standard deviation,

$$\bar{T} = \frac{1}{N}\sum_{k=1}^{N} T_k, \qquad \sigma_T = \left[\frac{1}{N}\sum_{k=1}^{N}(T_k - \bar{T})^2\right]^{1/2}, \quad N = 20. \tag{6}$$

The standard deviation is the ripple metric used for the comparison. At zero current it measures cogging, the torque variation caused by the permanent magnets and slots without applied phase current.

For a scalar statistic q, the empirical L3 uncertainty interval is

$$q_3 \;\pm\; e_q, \qquad e_q = \max\{|q_2 - q_1|, |q_3 - q_2|\}. \tag{7}$$

We use the same construction pointwise for waveforms. This interval is a refinement diagnostic, not a proven error bound or a statistical confidence interval. It does not cover reference-model error or angular quadrature error. The loaded means change direction between the last two increments, so a last-increment estimate alone would be particularly weak.

5 Forward reproduction results

5.1 Mean agreement and ripple disagreement

Table 1 separates the loaded mean from ripple. The mean is 0.15% low at half current and 0.16% low at full current. The respective reference means lie 0.05% and 0.07% beyond the empirical L3 intervals, so even the mean comparison is not agreement within that estimated spatial error. The zero-current mean is close to zero and is reported without a relative error.

Table 1: Finest-mesh comparison with saved reference output. Means and standard deviations are in N m per sector; deviations use the reference as denominator. Relative zero-current mean error is omitted.

a	$\bar{T}_{\mathrm{L3}}$	$\bar{T}_{\mathrm{ref}}$	$\Delta\bar{T}$ (%)	σ_{L3}	σ_{ref}	$\Delta\sigma$ (%)
0	0.00001	0.00010	—	0.00760	0.01071	-29.1
0.5	0.66106	0.66208	-0.15	0.04279	0.03895	+9.9
1	1.44023	1.44248	-0.16	0.08829	0.07935	+11.3

Ripple differs much more: 9.9% and 11.3% above the reference under load, and 29.1% below it at zero current. Figure 2 shows that the discrepancy is a waveform difference, not simply a mean shift. The reference lies outside the pointwise L3 intervals at 18, 18 and 17 of the 20 angles for zero, half and full current, respectively.

5.2 Spatial refinement

The maximum pointwise changes from L2 to L3 are 2.85×10^{-4}, 5.60×10^{-4} and 1.16×10^{-3} N m at the three currents. They are smaller than the corresponding differences from the reference, which reach 0.00715, 0.01471 and 0.02719 N m. Figure 3 shows decreasing waveform increments while the ripple discrepancies persist. The three ripple intervals are $[0.00733, 0.00786]$, $[0.04215, 0.04342]$ and $[0.08703, 0.08955]$ N m, each excluding its reference value.

On the ten even angles used in the reference optimization, the L3 ripple deviations are -33.8%, $+9.5\%$ and $+10.7\%$ when compared with the corresponding saved history values. The mismatch

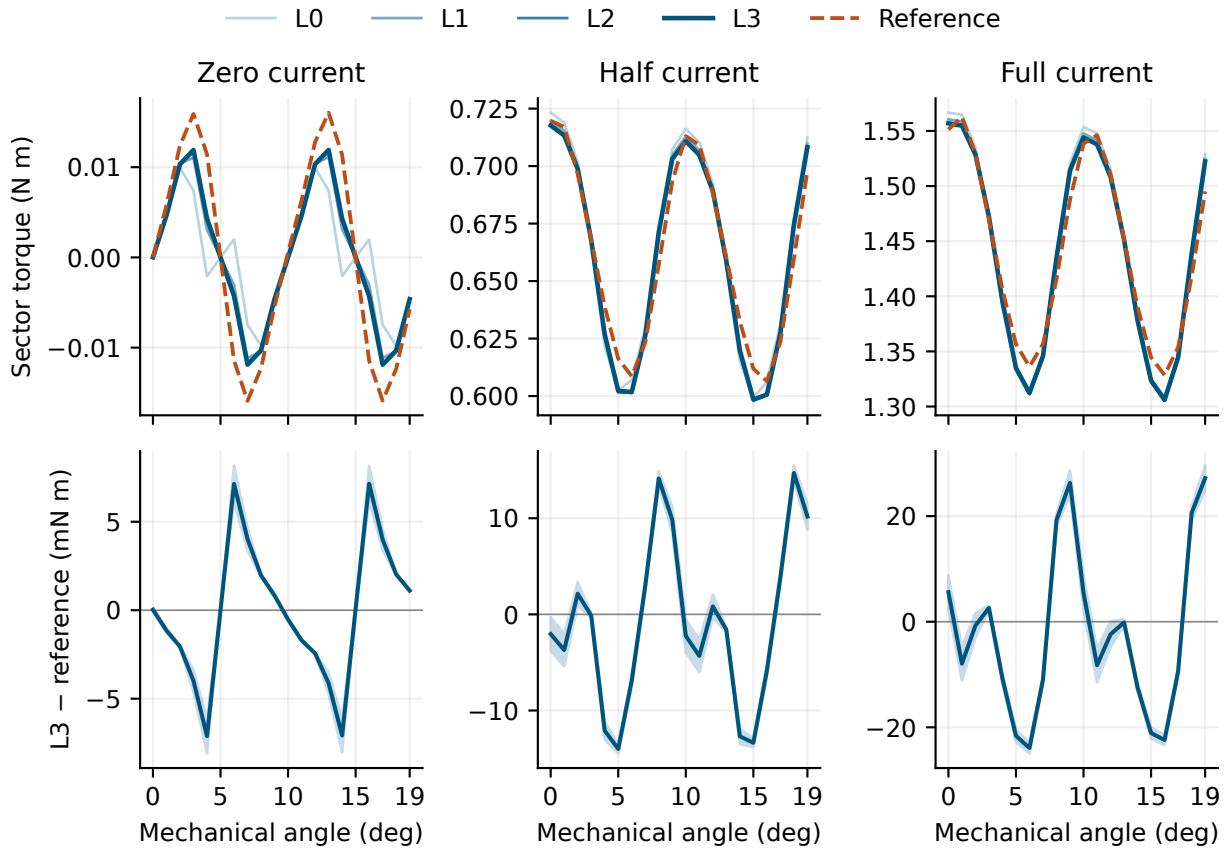


Figure 2: Top: GEODE torque waveforms at four spatial resolutions and the saved reference. Bottom: L3 minus reference, with the pointwise interval from Eq. (7). The three residual panels have independent vertical scales. The shaded regions estimate spatial change in GEODE; they do not quantify the reference's uncertainty.

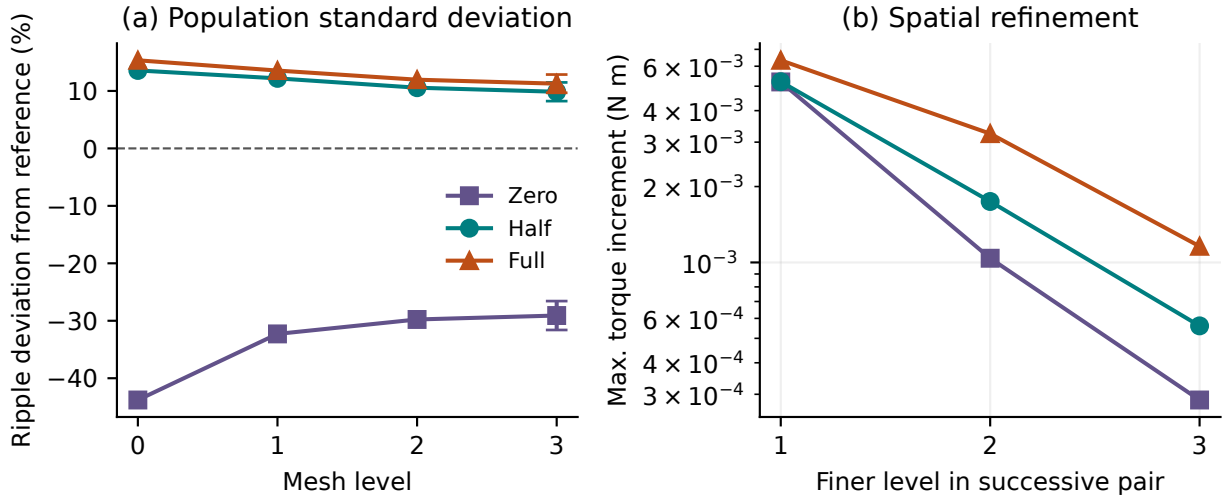


Figure 3: Spatial refinement of the initial motor. Left: relative ripple deviation, with L3 error bars from Eq. (7). Right: maximum waveform difference between successive meshes. Neither plot establishes continuum convergence of the independent reference calculation.

therefore persists on that sampling grid. This comparison is not an angular convergence study; denser and shifted angle grids remain necessary for design validation.

5.3 Sensitivity and unresolved causes

The controlled changes in Table 2 are too small to explain the observed waveform gap at the tested settings. The near-zero material difference has particularly little effect on torque despite its 6.66% local reluctivity difference. This is a measured sensitivity of the initial design and does not establish insensitivity after optimization.

Table 2: Maximum absolute torque change relative to the baseline at the same mesh level. Entries summarize the recorded angle/current subsets.

Change	Levels	Max. $\lvert\Delta T\rvert$ (N m)
Reference low-field interpolation and clamp	L2, L3	2.1×10^{-6}
Relative Newton tolerance 10^{-12} vs 10^{-8}	L2; L3 full current	7.3×10^{-8}
Exact winding areas vs mesh areas	L2	3.9×10^{-4}
Inner or outer half of the Arkkio band	L3	7.3×10^{-5}

Maxwell stress sampled on a single circle is less stable for the piecewise-constant P1 field. At L3 its maximum differences from the gap-averaged estimator range from about 0.0023 to 0.025 N m across the sampled radii and currents. That check does not independently validate Arkkio torque to high precision; the half-band and spatial-refinement results provide the present support for using the averaged estimator.

Two explanations remain open. First, the reference's eight harmonic orders restrict what the rotor and stator exchange through the air gap. Our conforming interface admits all modes resolved by the mesh. Filtering the already-solved L3 gap field at order 45 changes the diagnostic cogging standard deviation from about 0.0075 to 0.0184 N m. This demonstrates sensitivity to discarded modes, but it neither reproduces the reference value nor supplies the field that a truncated-coupling solve would produce. A matched harmonic-coupling solve is required to test the explanation.

Second, the saved reference uses degree-two isogeometric elements with 3×3 subdivisions per patch. It is a reproduction target, not a converged reference. Its own spatial and interface errors have not been bounded. The reference's cogging symmetry residual is approximately 5.5×10^{-4} N m, compared with less than 4.8×10^{-5} N m in GEODE L3; symmetry alone cannot detect a symmetric discretization error. The present evidence does not isolate either coupling truncation or reference coarseness as the cause.

5.4 Forward cost

Table 3 reports recorded local, sequential release-build timings on Apple Silicon. They describe the forward workload only, with no matched timing of the reference implementation. Warm angle steps typically require two or three Newton iterations; the initial L3 solves require 12–14. The full sensitivity study records 500 solves and 3491 seconds of summed per-solve wall time. Hardware identification, thread limits and repeat-run variation must be fixed before using timings for an optimizer comparison.

6 Planned gradient and optimization comparison

The following experiment is prospective. The forward results above do not include a motor shape derivative, an optimized geometry or a measured optimization speedup. Resolving the ripple comparison, or quantifying the remaining model difference, precedes those comparisons.

Table 3: Recorded baseline forward cost. Median wall time is taken over all 60 solves per mesh, including the three initial solves. Totals sum per-solve wall time and exclude mesh generation and setup.

Mesh	Nodes	Median (s)	Sum, 60 solves (s)
L0	3,153	0.09	6.1
L1	9,937	0.54	35.5
L2	36,083	3.32	222.2
L3	139,268	15.97	1072.8

For a design vector p and a converged reduced state $u(p)$ satisfying $R(u, p) = 0$, a scalar objective F has total derivative

$$\frac{\mathrm{d}F}{\mathrm{d}p} = F_p - \lambda^T R_p, \qquad R_u^T \lambda = F_u^T. \tag{8}$$

This identity follows by differentiating the residual and eliminating $\mathrm{d}u/\,\mathrm{d}p$. Implementing it requires derivatives of geometry, loads, material terms, constraints, interface motion and the actual torque functional. Access to the converged tangent is one ingredient, not a verification of all those derivatives. Directional finite-difference step sweeps and Taylor remainder tests must be run at initial and perturbed designs, with solver tolerances tightened enough to separate derivative error from state error.

The proposed objective is the sum of squared ripple over zero, half and full current, normalized by a fixed positive torque scale T_*:

$$F(p) = \sum_{a \in \{0, 0.5, 1\}} \frac{\sigma_T(p, a)^2}{T_*^2}. \tag{9}$$

This avoids division by the nearly zero mean cogging torque. Loaded mean torque, minimum air gap, bridge thickness, magnet area and valid geometry are constrained. The initial comparison fixes the outer envelope, axial length and control policy. These choices define a controlled shape study; they are not identical to every objective and parameter choice in the reference optimization.

Two experiments must be kept separate. The first compares optimizers within a common design space: exact gradients, finite differences as a diagnostic, competitive derivative-free search, and calibrated physical or waveform-surrogate models. The second expands a small engineering parameter basis with shape controls while preserving the constraints and validation target. Adding variables without a resolved performance gain would not demonstrate useful shape freedom.

Costs begin with the first model or calibration evaluation and include mesh updates, assembly, nonlinear solves, derivative assembly, factorizations, constraints, line searches and surrogate training. Each operating state and independent scalar adjoint output is counted; there is no assumption that one backward solve handles all angles and constraints. Methods share hardware, thread limits, initial designs, targets and budgets, with valid reuse available to each method. Candidates are checked on finer meshes, denser and shifted angles, tighter tolerances and withheld currents. A method that fails to attain the target within its budget is reported as such, rather than assigned a finite speedup ratio.

7 Discussion and limitations

The comparison demonstrates why forward validation should follow the optimization objective. Agreement in mean torque supports the excitation and normalization choices, but it is insufficient for an objective built from angular variation. The ripple discrepancy is larger than the observed

GEODE spatial changes and the tested material and solver sensitivities. Its cause still needs an experiment that changes the coupling formulation while holding the other inputs fixed.

The proposed benchmark remains two-dimensional and magnetostatic. It does not qualify thermal behavior, demagnetization, mechanical fatigue, three-dimensional end effects, losses or a complete efficiency map. The model is reconstructed from published computational artifacts; there is no new experimental validation against a physical motor. Only the initial design and the three specified current levels are evaluated here. Neither the material sensitivity nor the refinement behavior can be assumed to remain unchanged for a new geometry.

A simpler model is an essential control. At one fixed geometry, the existing linear slotless fixture is represented by a three-coefficient sinusoid with held-out-angle error near numerical precision (Appendix A). That observation does not test unseen designs, but it shows why parameter count alone is a poor reason to expect a gradient advantage. A competitive nonlinear reduced model might also capture the design directions needed for the slotted motor.

8 Conclusion

A nonlinear finite-element reconstruction matches the published initial motor's loaded mean torque within 0.16%, while its ripple differs by 9.9–11.3% under load and 29.1% at zero current. Four spatial resolutions and targeted sensitivity checks expose the discrepancy but do not resolve its cause. The resulting model and evidence support the next comparison: whether controlling interface and discretization error makes torque ripple a stable target for shape optimization. The benefit of full-field gradients remains an experimental question.

Code and data availability

The evidence used here is fixed at GEODE commit `581ee7ad`. The repository contains the raw motor runs and analysis, geometry and mesh-generation scripts, and solver verification logs. Reference data are extracted from the authors' ICEM snapshot at `b145aaf4`, a later repository revision rather than a claim to the original conference release. Source hashes, extraction checks and the upstream GPL-3.0 license are retained. The paper package includes a manifest, copied figure inputs and a regeneration script; it does not require rerunning FEM to rebuild the figures. Recomputing the physical study requires the separately documented mesh and solver commands.

A Linear slotless screening control

The stored slotless fixture has 24 distinct angles. Fitting $T(\phi) = c_0 + c_1 \cos\phi + c_2 \sin\phi$, with electrical angle $\phi = p\theta_r$ and pole-pair count p, on alternate angles and evaluating on the other twelve gives normalized held-out RMSE 3.08×10^{-10} and maximum error 6.83×10^{-10}, using the recorded analytic-amplitude normalization. This is a locked-rotor angle sweep, not synchronous operating ripple. For a fixed linear field operator and sinusoidal sources, the field lies in a small basis of source responses; a quadratic torque functional can contain only the corresponding finite set of harmonics. The particular fixture needs just the three fitted coefficients. This is an angular interpolation test at one geometry, not a shape-gradient test or evidence of generalization across designs.

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